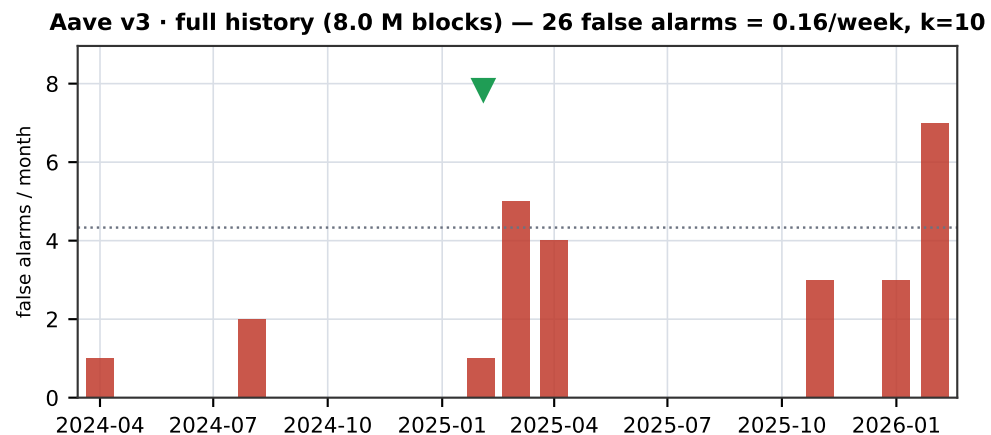
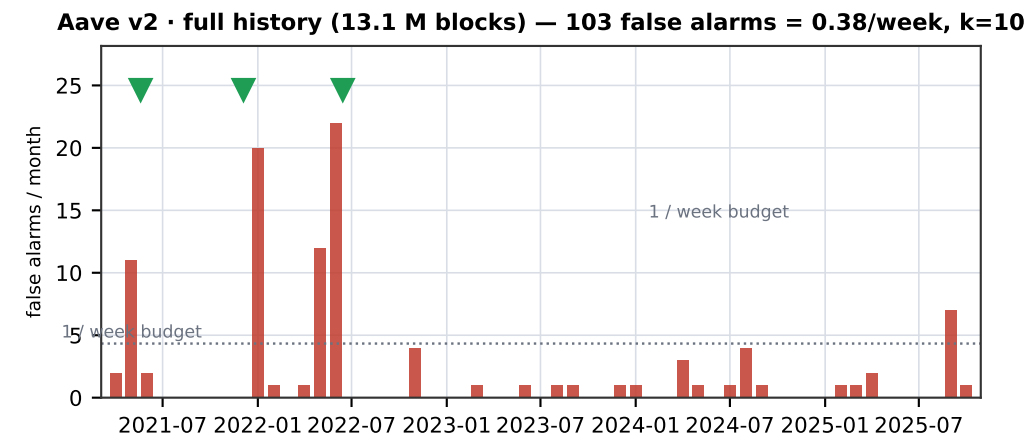
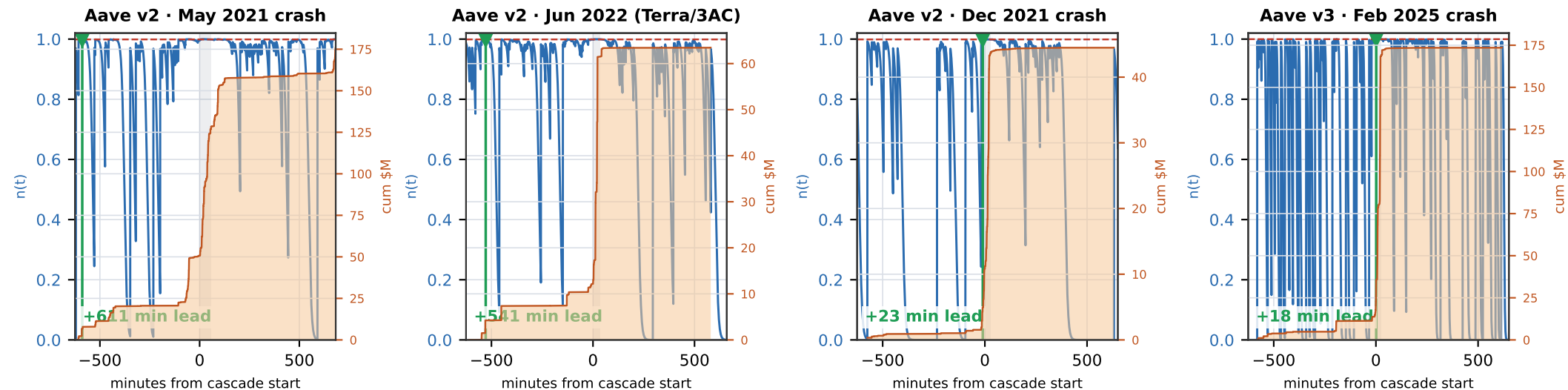


Hawkes branching-ratio alarm: early on cascades (top), quiet otherwise (bottom)



— $n(t)$ branching ratio - - - alert threshold ▼ alarm fire / cascade detected — cumulative liquidation \$ ■ false alarms / month